ELEC50001

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2026

CIRCUITS AND SYSTEMS

Tuesday 12 May 2026, 2:30 PM

Time allowed: 2:00 hours

There are THREE questions on the paper.

Answer ALL questions.

Question 1 carries a weighting of 50%. The remaining questions carry a Weighting of 25% each.

Any special instructions for invigilators and information for Candidates are on page 1.

Examiners responsible First Marker(s): P.Y.K. Cheung

Second Marker(s): Aaron Zhao

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Information for Candidates:

The following notation is used in this paper:

1. Unless explicitly indicated otherwise, digital circuits are drawn with their inputs on the left and their outputs on the right.
2. Within a circuit, signals with the same name are connected even if no connection is shown explicitly.
3. The notation X[2:0] denotes the three-bit number X2, X1 and X0. The least significant bit of a binary number is always designated bit 0.

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1. (a) *Figure 1.1* shows a simplified SPICE model for the MAX 40110 operational amplifier. The model has three parts: the input stage, the gain stage and the output stage. V+ and V- are the positive and negative power supplies to the operation amplifier respective. Based on the datasheet provided in the Appendix, answer the following questions:

(i) For the input stage, determine the values of R1, C1 and V1.

[3]

(ii) For the gain stage, determine the values of R2, C2 and the transconductance of G1.

[3]

(iii) The MAX 40110 is connected as a unity gain amplifier with V+ at 3.3V and V- at 0V. Sketch the output signal VOUT if the input signal VIN is an ideal square wave signal at a frequency of 1MHz with low voltage at 0V and high voltage at 3.3V.

.

[4]

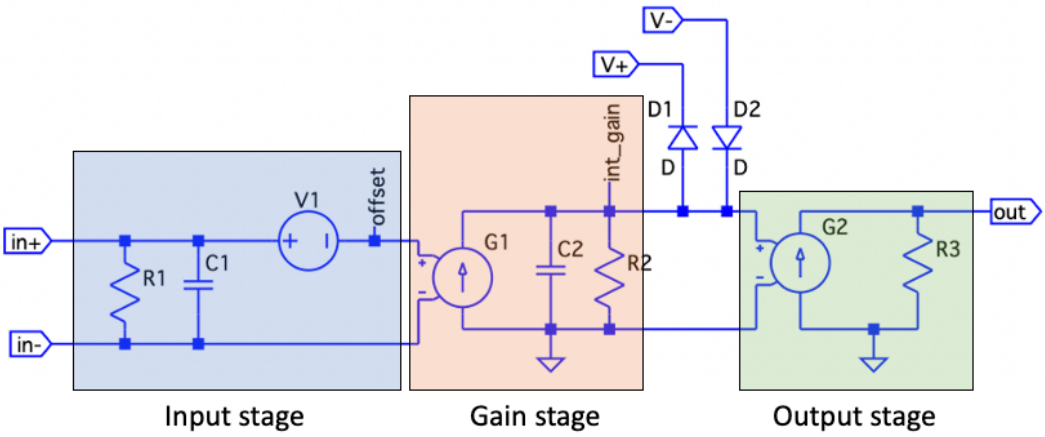


Figure 1.1

SOLUTION TO Q1 (a)

This question tests students understand of the datasheet of an operational amplifier and how to derive the macro model SPICE model for the op-amp.

(i) V1 = input offset voltage **V_{OS} = 0.14mV**. This is the maximum at 25 deg C.
C1 = input capacitance **C_{IN} = 2.5pF**.
R1 = input resistance = input voltage/input bias current and needs to be derived.
Input voltage = VDD/2 = 1.65V. Input bias current is 12pA (up to 80 deg C).
Therefore **R1 = 137.5 x 10^9 ohms**.

[3]

(ii) GBP = 15MHz. Open-loop gain A_{OL}=110dB or 400,000.

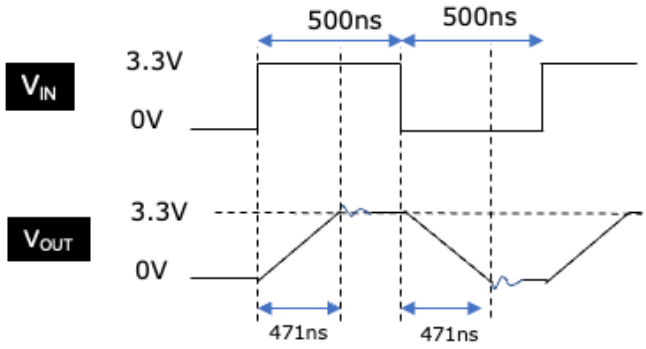
Dominant pole $f_p = \frac{GBP}{A_{OL}} = \frac{1.5\ x\ 10^7}{400000} = 37.5Hz = 1/2\pi R_2 C_2$.

Assume $\boldsymbol{C_2 = 1nF}$, **R2 = 4.25MΩ**.

$gm \times R_2 = A_{OL}$, hence $gm = \frac{400{,}000}{4.25} \times 10^{-6} = \mathbf{0.042}$.

[3]

(iii) Slew rate is 7V/us. Therefore 3.3V would take 0.47us to traverse. Hence the output signal is:



[4]

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(b) (i) It is known that $1 + x^3 + x^{10}$ is a primitive polynomial, design in SystemVerilog a 10-bit pseudo-random binary sequence generator module with the interface signals defined in *Figure 1.2*.

[6]

(ii) For your design, write down the first 5 values of the sequence.

[4]

```
module prbs_10(
  input logic           clk,
  input logic           rst,
  output logic [9:0]    data_out
);
```

Figure 1.2

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SOLUTION TO Q1 (b)

This question tests students understand pseudo-random binary sequence generator, primitive polynomial, basic SystemVerilog and the need to initialise the generator to a valid state.

(i)

```
module prbs_10(
  input logic           clk,
  input logic           rst,
  output logic [9:0]    data_out
);

  logic [10:1]     sreg;

  always_ff @ (posedge clk, posedge rst)
  begin
    if (rst)
      sreg <= 10'b1;
    else
      sreg <= {sreg[9:1], sreg[10] ^ sreg[3]};
  end
  assign data_out = sreg;

endmodule
```

[6]

(ii) Assuming that the starting state is 10'h001, then the sequence of 5 values are: {10'h001, 10'h002, 10'h004, 10'h00A, 10'h012}.

[4]

(c) *Figure 1.3* shows a pipelined circuit with three flip-flops: FF1, FF2 and FF3, and two combinational logic circuit modules: P and Q. FF1 and FF3 are falling-edge triggered while FF2 is rising-edge triggered. For all flip-flops, the clock-to-output delay is 0.5ns and the setup time is 1ns. The propagation delays of P and Q are in the range of 2 ns to 8 ns and 3 ns to 9 ns respectively. You may assume that the hold times of all flip-flops are zero.

(i) If the clock signal is symmetrical and has a mark-space ratio of 1:1, derive the maximum operating frequency of the circuit.

[6]

(ii) It is possible to increase the maximum operating frequency using a non-symmetric clock signal with mark-space ratios different from 1:1. What is the maximum operating frequency of the circuit and with what mark-space ratio allows the circuit to work at the highest possible clock frequency?

[4]

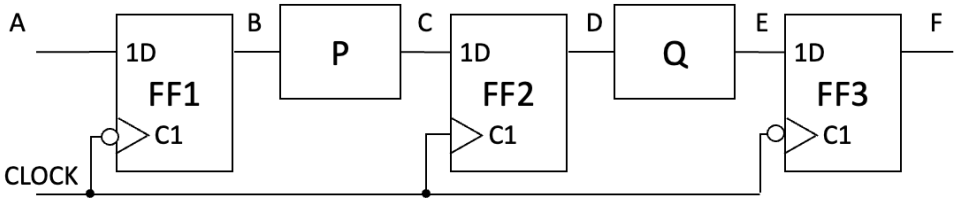
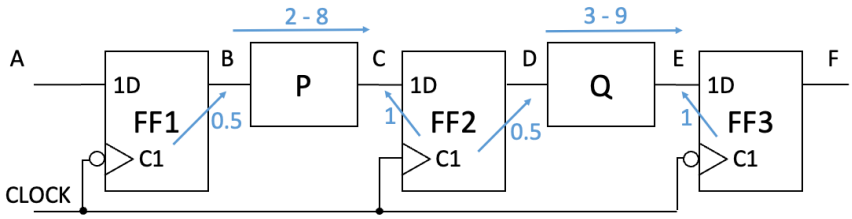


Figure 1.3

SOLUTION TO Q1 (c)

This question examines student's understanding of digit circuit timing constraints in a pipelined system. It further tests student's ability to use asymmetrical clock signal to maximum operating frequency.



(i) For FF2 data input, the constraint is:

$$T_{cq_1} + t_{p_P}(\max) + t_{s_2} \leq \frac{T_{clk}}{2}.$$

Therefore. $0.5 + 8 + 1 = 9.5 \leq \frac{T_{clk}}{2}$, $T_{clk} \geq 19$ns.

For FF3 data input, the constraint is:

$$T_{cq_2} + t_{p_Q}(\max) + t_{s_3} = 10.5 \leq \frac{T_{clk}}{2}.$$

Therefore $0.5 + 9 + 1 \leq \frac{T_{clk}}{2}$, $T_{clk} \geq 21$ns.

FF3 data input dictates the max frequency = 47.6MHz.

[6]

(ii) If the clock signal has a low width (space) of 9.5ns and a high width (mark) of 10.5 ns, all setup timing constraints are satisfied. Clock period is 20ns.

Therefore, max frequency = 50MHz.
The mark-space ratio will be 10.5:9.5 ≈1.1 : 1.

[4]

(d) *Figure 1.4* shows a 4k x 4 RAM chip with address bus ADR, data bus DATA, a write enable signal WR, a chip select signal CS and a synchronising clock CLK.

(i) Specify the size of the address and data buses for the RAM chip.

[2]

(ii) An 8k x 8 RAM module is required in a microprocessor system with a 16-bit address bus and an 8-bit data bus. The starting address of the RAM module is 16'h8000.

a) Design, in the form of a schematic diagram, a circuit using four 4k x 4 RAM chips shown in *Figure 1.4* to implement the 8k x 8 RAM module.

[5]

b) Derive the Boolean expression for the chip select signals required for each of the 4k x 4 RAM chip.

[3]

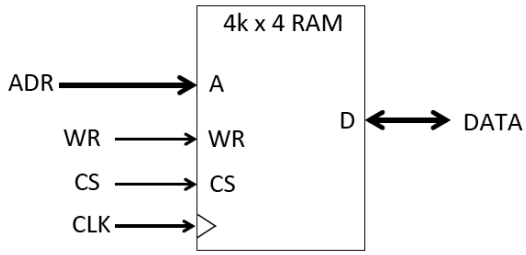


Figure 1.4

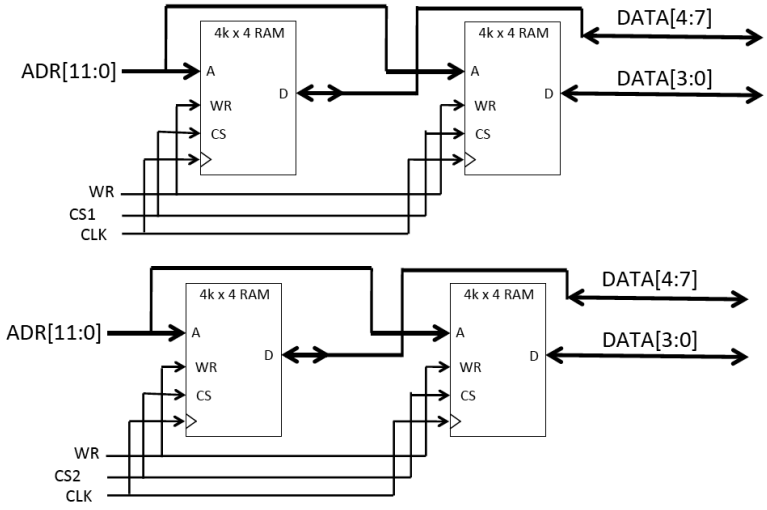
SOLUTION TO Q1 (d)

This question tests student's understanding of basic memory organisation and address decoding.

(i) Address: 12 bits, Data 4 bits..

[2]

(ii) a)



[5]

b) CS1 address range: 16'h8000 to 16'h8fff.
CS2 address range: 16'h9000 to 16'h9fff.
Therefore:

CS1 = A15 & !A14 & !A13 & !A12
CS2 = A15 & !A14 & !A13 & A12

[3]

(e) (i) Explain the principle of operation of a 12-bit pulse-width modulation (PWM) digital-to-analogue converter (DAC).

[4]

(ii) Design in SystemVerilog a PWM module with an interface as defined in *Figure 1.5*.

[6]

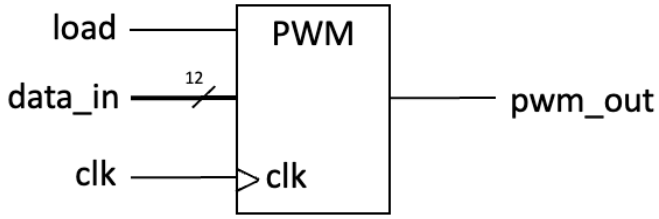


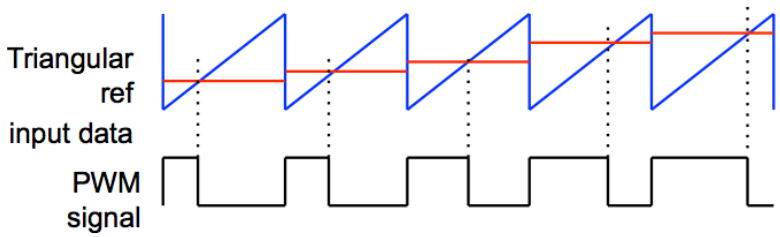
Figure 1.5

SOLUTION TO Q1 (e)

This question tests student's understanding PWM based DAC.

(i) Bookwork.

Generate a triangular signal (in the form of a 12-bit counter) and compare the input value *data_in* to that of the counter value. Set *pwm_out* to be high if the counter value is lower than *data_in*, otherwise set it to low. The PWM output is then lowpass filtered to obtain the analogue output signal.



[4]

(ii)

```
module pwm # (parameter WIDTH = 12)(
    input  logic                clk,        // system clock
    input  logic [WIDTH-1:0]    data_in,    // input data for conversion (default 12 bits)
    input  logic                load,       // high pulse to load new data
    output logic                pwm_out     // PWM output
);
    logic [WIDTH-1:0]       d;              // internal register
    logic [WIDTH-1:0]       count;          // internal 1WIDTH-1 bit counter

    always_ff @ (posedge clk)
        if (load == 1'b1) d <= data_in;

    initial count = {WIDTH{1'b0}};          // init count FPGAs

    always_ff @ (posedge clk) begin
        count <= count + 1'b1;              // sawtooth signal

        if (count >= d)                     // comparator
            pwm_out <= 1'b0;
        else
            pwm_out <= 1'b1;
        end

endmodule
```

[6]

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2. *Figure 2.1* shows a finite state machine (FSM) clocked by the signal *clk,* with input signal *in* and 2-bit output signals *out[1:0]*. The FSM is implemented using an 8 x 4-bit ROM and a 4-bit register. The input *in* is connected to the upper address bit of the ROM *ADR[2]*. The output of the ROM *D[3:0]* is connected to the data input of the register. The lower 2 bits of the ROM data *D[1:0]* provide the next state value, which are fed back to the lower 2 bits of the ROM address *ADR[1:0]* through the state register *s[1:0]*. The upper 2 bits of the ROM data *D[3:2]* provide the FSM output values *out[1:0]*. The reset input *rst* clears the register.

(a) Determine the contents of the ROM to implement the FSM according to the state diagram shown in *Figure 2.2*.

[10]

(b) The same FSM in (a) is to be implemented using Altera's MAX10 FPGA device with Logic Elements (LEs) instead of ROM. Each LE consists of a 4-input lookup table and a register. Estimate, with justifications, how many LEs are required to implement the FSM using *one-hot state encoding*. State any assumptions used.

[5]

(c) Design this state machine in SystemVerilog.

[10]

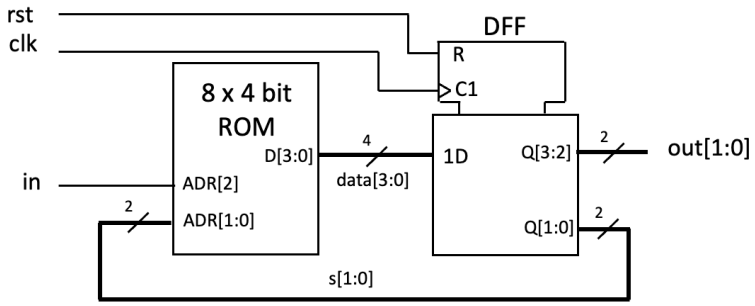


Figure 2.1

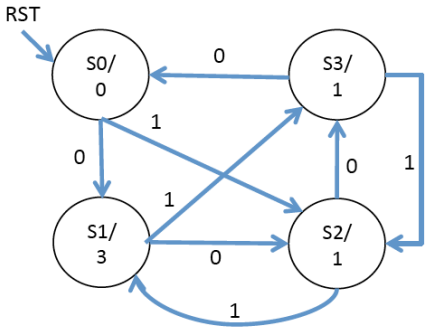


Figure 2.2

SOLUTION TO Q2

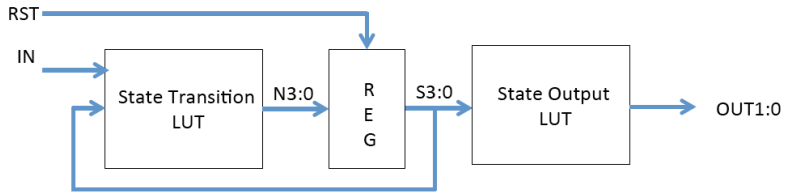
This question tests student's ability to implement finite state machine using two technologies: regular logic blocks such as ROM and registers, and using FPGAs. It also test student's ability to specify FSM using SystemVerilog.

(a)

ROM Address ADR[2:0] {in, s[1:0]}	ROM Data D[3:0]
0 00	11 01
0 01	01 10
0 10	01 11
0 11	00 00
1 00	01 10
1 01	01 11
1 10	11 01
1 11	01 10

. [10]

(b) The FPGA implementation with one-hot encoding would look like this.



Since each LE can implement arbitrary logic with 4 inputs, the state transition logic will need 4 LEs, and the output logic needs 2 LEs. Total LE = 6.

. [5]

(c)

```
module FSM (
    input  logic     rst,    // reset input
    input  logic     clk,    // clock
    input  logic     in,     // input signal
    output logic [1:0] out   // output signals
);
    // define states
    typedef enum {S0, S1, S2,S_3} my_state;
    my_state   current_state, next_state;

    // state registers
    always_ff @(posedge clk)
        if (rst)    current_state <= S0;
        else        current_state <= next_state;

    // next state logic
    always_comb
        case (current_state)
            S0: if (in==1'b1) next_state = S2;
                else next_state = S1;
            S1: if (in==1'b1) next_state = S3;
                else next_state = S2;
            S2: if (in==1'b1) next_state = S1;
                else next_state = S3;
            S3: if (in==1'b1) next_state = S2;
                else next_state = S0;
        default: next_state = current_state;
        endcase

    // output logic
    always_comb
        case (current_state)
            S0: out = 2'b0;
            S0: out = 2'b1;
            S0: out = 2'b1;
            S0: out = 2'b11;
        default: out = 2'b0;
        endcase
endmodule
```

. [10]

3. (a) *Figure 3.1* shows an integrator circuit implemented using a dual supply operational amplifier. Show that the transfer function for this integrator is given by:

$$\frac{V_{out}(s)}{V_{in}(s)} = -\frac{100}{s}$$

[5]

(b) *Figure 3.2* shows a summing integrator with two input voltages X and Y. Note that the input resistors are now 100kΩ instead of 10kΩ. Show that the relationship between the output V_1 and the inputs X and Y is given by the following:

$$V_1(s) = -\frac{2}{1 + 0.2s}\,(X(s) + Y(s))$$

[10]

(c) *Figure 3.3* shows a filter circuit combining the circuits from (a) and (b) above and feedback. Show that the transfer function of this circuit is that of a second-order low pass filter of the form:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = -\frac{1000}{s^2 + 5s + 1000}$$

[10]

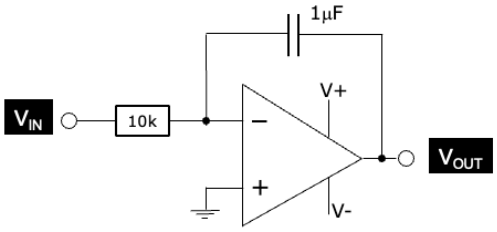


Figure 3.1

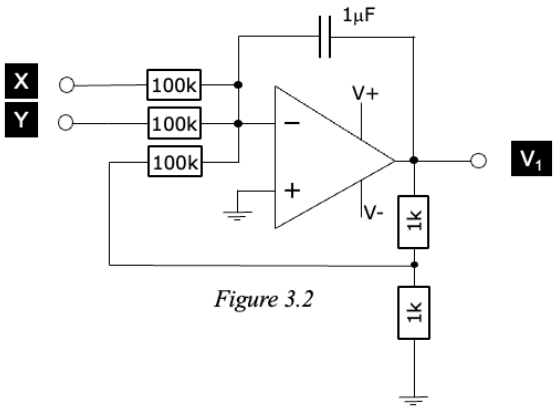


Figure 3.2

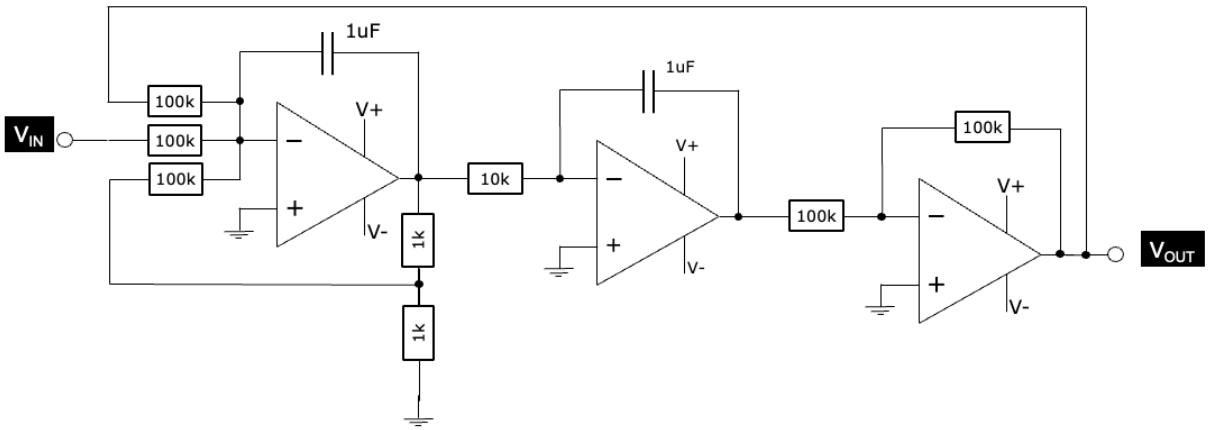


Figure 3.3

SOLUTION TO Q3

This question tests student's understanding application of op-amp from first principle.

(a) Impedance of capacitor is 1/sC. Current to virtual earth is $\frac{V_{IN}}{R}$. Therefore $V_{OUT} = -\frac{1}{sRC}V_{IN}$.

Substitute values for R and C gives:

$$K(s) = \frac{V_{out}(s)}{V_{in}(s)} = -\frac{100}{s} \qquad \text{Eq. 1}$$

[5]

(b) The integrator now has three input branches. Using principle of superposition, we can consider these separately.

For X: $V_1(s) = -\frac{1}{RC}\frac{1}{s}X(s) = -\frac{10}{s}X(s)$

For Y: $V_1(s) = -\frac{1}{RC}\frac{1}{s}Y(s) = -\frac{10}{s}Y(s)$

For lower branch: $V_1(s) \approx -\frac{1}{RC}\frac{1}{s}\frac{V_1(s)}{2} = -\frac{5}{s}V_1(s)$

Therefore, total $V_1(s)$ is sum of these three terms:

$$V_1(s) = -\left(\frac{10}{s}X(s) + \frac{10}{s}Y(s) + \frac{10}{s}V_1(s)/2\right) \qquad \text{Eq. 2}$$

Hence, $$V_1(s) = -\frac{2}{1+0.2s}\,(X(s) + Y\,(s)) \qquad \text{Eq. 3}$$

[10]

(c) This filter architecture is known as state variable filter. From Eq. 1:

$$X(s) = V_{OUT}(s) = \frac{100}{s}V_1(s) \qquad \text{Eq.4}$$

$$Y(s) = V_{IN}(s) \qquad \text{Eq. 5}$$

Substitute Eq. 4 and 5 into Eq. 3 and eliminate $V_1(s)$ gives

$$V_{OUT}(s) = -100\frac{1}{s}\left\{\frac{2}{1+0.2s}\left(V_{OUT}(s) + V_{IN}(s)\right)\right\}$$

$$= -\left\{\frac{1000}{5s+s^2}\left(V_{OUT}(s) + V_{IN}(s)\right)\right\}$$

Therefore:

$$V_{OUT}(s)\left(1 + \frac{1000}{5s+s^2}\right) = -\frac{1000}{5s+s^2}V_{IN}(s)$$

$$H(s) = \frac{V_{OUT}(s)}{V_{IN}(s)} = -\frac{1000}{s^2+5s+1000}$$

[10]

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APPENDIX

Click here to ask about the production status of specific part numbers.

MAX40110/MAX40111 — 15MHz, Low-Noise, Precision Operational Amplifiers

General Description

The MAX40110/MAX40111 offer a unique combination of high speed, precision, low noise, and low-voltage operation, making them ideally suited for a large number of signal processing functions including filtering and amplification of signals in portable and industrial equipment.

The MAX40110/MAX40111s' rail-to-rail input/outputs and low noise guarantee maximum dynamic range in demanding applications such as 12- to 14-bit SAR ADC drivers. Unlike traditional rail-to-rail input structures, input crossover distortion is absent due to an optimized input stage with an ultra-quiet charge pump. The input offset voltage is as low as 30µV due to the auto-calibration performed on power-up. The devices also include a fast-power-on shutdown mode for further power savings.

The MAX40110/MAX40111 operate from a supply range of 1.8V to 5.5V over the -40°C to +125°C extended operating temperature range and consume only 750µA (typ) supply current. The MAX40110 is offered in an ultra-thin 6-µDFN package, and the MAX40111 is available in a tiny 6-bump wafer-level-package (WLP) with a 0.4mm pitch.

The MAX40110/MAX40111 are an extension of the MAX44259/60/61/63 and MAX44290 families with a reduced power up time of less than 1ms.

Applications

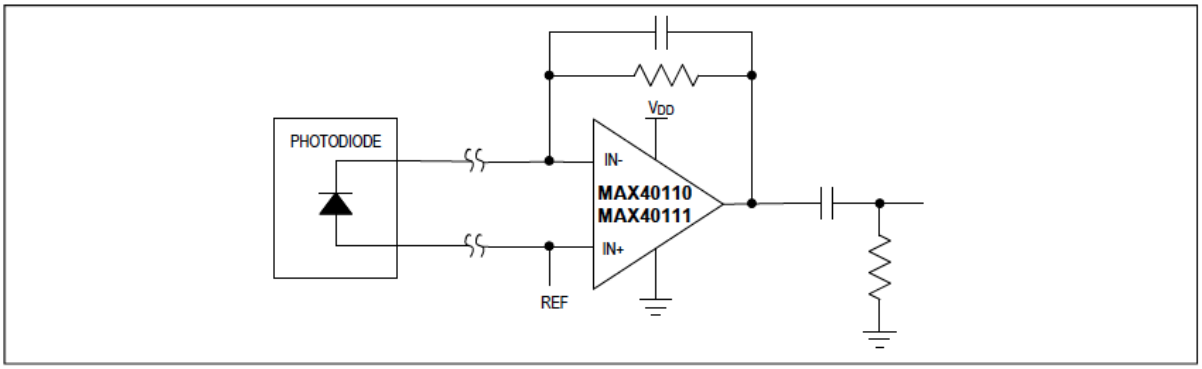
- Wearable Devices
- Home Medical (Blood Glucose, Weight Scale, Blood Pressure, EKG)
- Industrial IOT (Pressure, Flow, Level, Temperature, Proximity)
- Optical Sensors
- SAR ADC Driver

Benefits and Features

- Fast Power-up Time of less than 1ms
- Supply Voltage Range: 1.7V to 5.5V
- Low 140µV (Max) Input Offset Voltage with Power-On Auto-Cal at +25ºC
- 15MHz GBW
- Low 12.7nV/√Hz Input Voltage-Noise Density
- 750µA Quiescent Current
- Rail-to-Rail Inputs and Outputs
- Internal EMI Rejection
- <1pA Low Input Bias Current
- Power-Saving Shutdown Mode
- Available in a Tiny 6-bump WLP and a Thin 6-µDFN Package

Ordering Information appears at end of data sheet.

Typical Application Circuit



maxim integrated™

19-101187; Rev 0; 7/21

Absolute Maximum Ratings

V_{DD} to GND ..... -0.3V to +6V
SHDN ..... -0.3V to +6V
IN+, IN- Maximum Voltage ..... Self Limiting*
OUT to GND ..... -0.3 to (V_{DD} + 0.3V)
Output Short-Circuit Duration to V_{DD} or GND ..... Continuous
Continuous Current into Any Input/Output Pin ..... ±10mA
Differential Input Voltage ..... ±6V
Continuous Power Dissipation (µDFN (Derate 2.1mW/°C above +70°C)) 170mW
Continuous Power Dissipation (WLP (Derate 10.51mW/°C above +70°C)) 840.8mW
Operating Temperature Range ..... -40°C to +125°C
Junction Temperature ..... +150°C
Storage Temperature Range ..... -65°C to +150°C
Lead Temperature (soldering, 10s) ..... +300°C
Soldering Temperature (reflow) ..... +260°C

Note: *Not to exceed +6V.

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Package Information

WLP

Package Code	N60T1+1
Outline Number	21-100568
Land Pattern Number	Refer to Application Note 1891
Thermal Resistance, Four-Layer Board:	
Junction to Ambient (θ_{JA})	95.15°C/W
Junction to Case (θ_{JC})	N/A

Electrical Characteristics

(V_{DD} = +3.3V, GND = 0, V_{CM} = V_{DD}/2, R_{LOAD} = 10kΩ to V_{DD}/2, $V_{\overline{SHDN}}$ = V_{DD}, T_A = -40°C to +125°C, unless otherwise noted. Typical values are at +25°C. (Note 1))

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
DC SPECIFICATIONS						
Input Common-Mode Range	V_{CM}	Guaranteed by CMRR test	-0.1		V_{DD} + 0.1	V
Common-Mode Rejection Ratio	CMRR	−0.1V ≤ V_{CM} ≤ V_{DD} + 0.1V	75	90		dB
Input Offset Voltage	V_{OS}	T_A = +25°C (Note 2)		30	140	µV
		-40°C ≤ T_A ≤ +125°C (Note 3)			190	
		-40°C ≤ T_A ≤ +125°C (Note 4)			600	

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Electrical Characteristics (continued)

(V_{DD} = +3.3V, GND = 0, V_{CM} = V_{DD}/2, R_{LOAD} = 10kΩ to V_{DD}/2, $V_{\overline{SHDN}}$ = V_{DD}, T_A = -40°C to +125°C, unless otherwise noted. Typical values are at +25°C. (Note 1))

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
Input Offset Voltage Drift	TC V_{OS}	(Note 5)			1	6	µV/°C
Input Bias Current	I_B	-40°C ≤ T_A ≤ +85°C (Note 5)			1	12	pA
		-40°C ≤ T_A ≤ +125°C (Note 5)				200	
Input Offset Current	I_{OS}	-40°C ≤ T_A ≤ +85°C (Note 5)			1	12	pA
		-40°C ≤ T_A ≤ +125°C (Note 5)				200	
Open-Loop Gain	A_{OL}	0.4V ≤ V_{OUT} ≤ (V_{DD} - 0.4V)	R_{LOAD} = 10kΩ to V_{DD}/2	100	115		dB
			R_{LOAD} = 600Ω to V_{DD}/2	91	100		
Output Voltage Swing High	V_{OH}	V_{DD} - V_{OUT}	R_{LOAD} = 10kΩ to V_{DD}/2			10	mV
			R_{LOAD} = 600Ω to V_{DD}/2			40	
Output Voltage Swing Low	V_{OL}	V_{OUT}	R_{LOAD} = 10kΩ to V_{DD}/2			20	mV
			R_{LOAD} = 600Ω to V_{DD}/2			50	
Short Circuit Current	I_{SC}	To V_{DD} or GND			50		mA
AC SPECIFICATIONS							
Input Voltage Noise Density	e_N	f = 10kHz			12.7		nV/√Hz
Input Voltage Noise		0.1Hz to 10Hz			10		$µV_{P-P}$
Input Current Noise Density	I_N	f = 10kHz			1.2		fA/√Hz
Input Capacitance	C_{IN}				2.5		pF
Gain Bandwidth Product	GBW				15		MHz
Slew Rate	SR	AV = 1V/V, V_{OUT} = $2V_{P-P}$, 10% to 90%			7		V/µs
Capacitive Loading Stability		No sustained oscillation			300		pF
Total Harmonic Distortion + Noise	THD+N	V_{OUT} = $2V_{P-P}$, A_V = 1V/V, R_{LOAD} = 10kΩ, f = 10kHz			-105		dB
Settling Time		V_{OUT} = $2V_{P-P}$, A_V = 1V/V, C_{LOAD} = 30pF, settle to 0.01%,			1.7		µs
Output Transient Recovery Time		ΔV_{OUT} = 0.2V, V_{DD} = 3.3V, A_V = 1V/V, R_S = 20Ω, C_{LOAD} = 1nF			1		µs
POWER SUPPLY							
Supply Voltage	V_{DD}	Guaranteed by PSRR		1.8		5.5	V
		T_A = 0°C to +70°C		1.7		5.5	
Power-Supply Rejection Ratio	PSRR	V_{CM} = V_{DD}/2		78	95		dB
Supply Current	I_{DD}				750	1,200	µA
Shutdown Supply Current	$I_{\overline{SHDN}}$					1	µA

Electrical Characteristics (continued)

(V_{DD} = +3.3V, GND = 0, V_{CM} = V_{DD}/2, R_{LOAD} = 10kΩ to V_{DD}/2, $V_{\overline{SHDN}}$ = V_{DD}, T_A = -40°C to +125°C, unless otherwise noted. Typical values are at +25°C. (Note 1))

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
Power-Up Time	t_{ON}	Includes power-on V_{OS} self-calibration	V_{DD} = 0V to 3V step, AV = 1V/V (Note 5)		0.6	1	ms
Shutdown Turn-On Time	$t_{\overline{SHDN}}$	-40°C < T_A < +125°C (Note 5)			15	29	µs
LOGIC INPUT DC CHARACTERISTICS ($\overline{SHDN}$)							
Input Low Level	V_{IL}					0.5	V
Input High Level	V_{IH}			1.3			V
Shutdown Input Bias Current	I_{IL} / I_{IH}					1	µA

Note 1: All devices are 100% production tested at T_A = +25°C. All temperature limits are guaranteed by design.

Note 2: At T_A = +25°C, on power-up and after calibration.

Note 3: For any temperature values between -40°C to +125°C, on power-up and after calibration.

Note 4: For any temperature values between -40°C to +125°C, it indicates the maximum drift from the initially calibrated value at power-up.

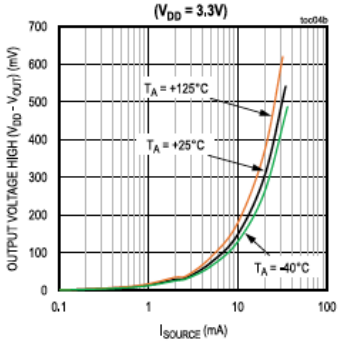
Note 5: Not production tested, guaranteed by design and bench characterization.

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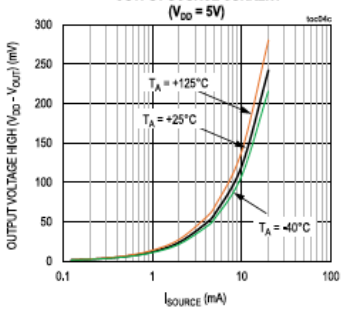
Typical Operating Characteristics (continued)

(V_{DD} = +3.3V, GND = 0, V_{CM} = V_{DD}/2, R_{LOAD} = 10kΩ to V_{DD}/2, $\overline{SHDN}$ = V_{DD}, T_A = +25°C, unless otherwise noted.)

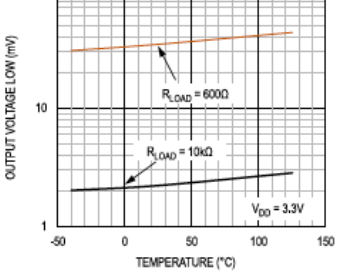
OUTPUT VOLTAGE HIGH vs. OUTPUT SOURCE CURRENT (V_{DD} = 3.3V)



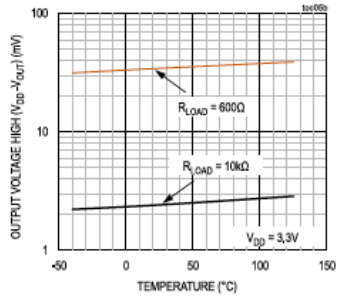
OUTPUT VOLTAGE HIGH vs. OUTPUT SOURCE CURRENT (V_{DD} = 5V)



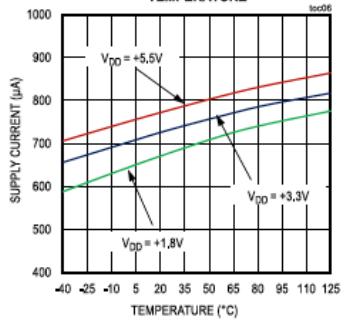
OUTPUT VOLTAGE LOW vs. TEMPERATURE



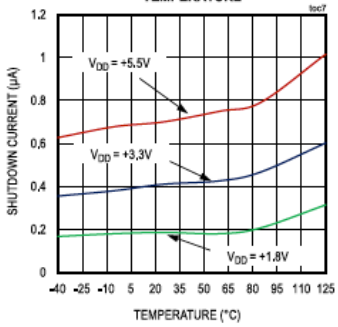
OUTPUT VOLTAGE HIGH vs. TEMPERATURE



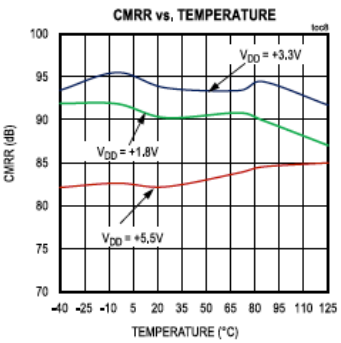
QUIESCENT SUPPLY CURRENT vs. TEMPERATURE



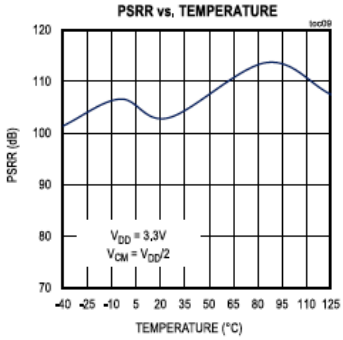
SHUTDOWN SUPPLY CURRENT vs. TEMPERATURE



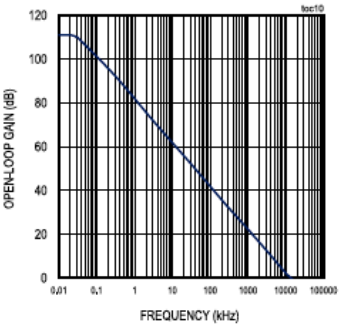
CMRR vs. TEMPERATURE



PSRR vs. TEMPERATURE



OPEN-LOOP GAIN vs. FREQUENCY

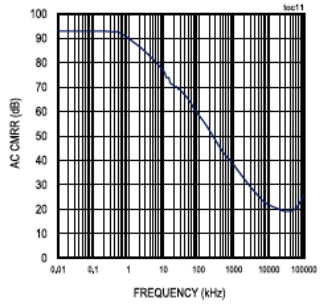


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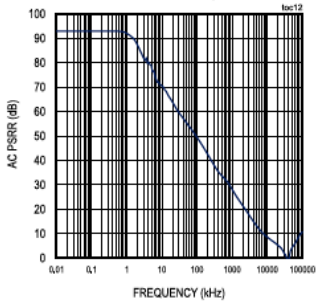
Typical Operating Characteristics (continued)

(V_{DD} = +3.3V, GND = 0, V_{CM} = V_{DD}/2, R_{LOAD} = 10kΩ to V_{DD}/2, $\overline{SHDN}$ = V_{DD}, T_A = +25°C, unless otherwise noted.)

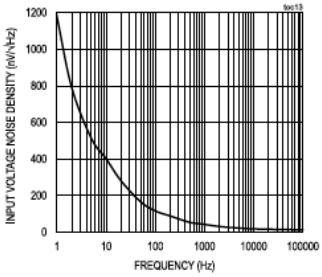
AC CMRR vs. FREQUENCY



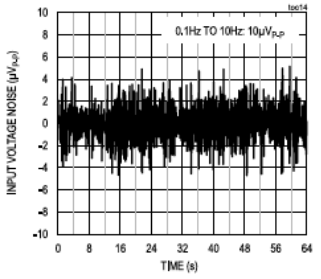
AC PSRR vs. FREQUENCY



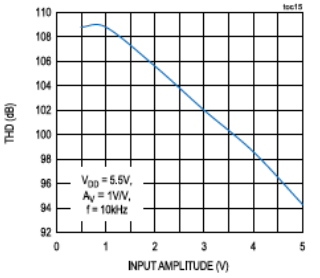
INPUT VOLTAGE NOISE DENSITY vs. FREQUENCY



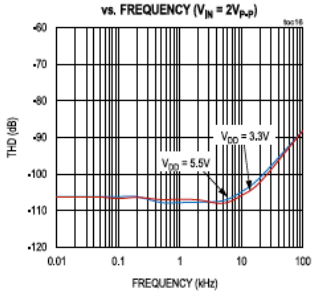
0.1Hz TO 10Hz INTEGRATED NOISE



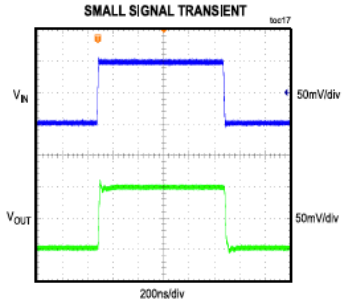
TOTAL HARMONIC DISTORTION vs. AMPLITUDE



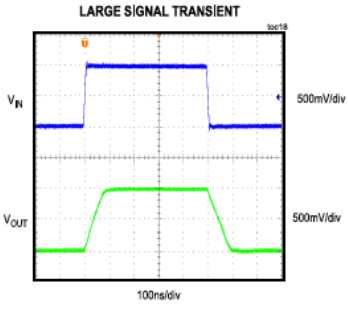
TOTAL HARMONIC DISTORTION vs. FREQUENCY (V_{IN} = $2V_{P-P}$)



SMALL SIGNAL TRANSIENT



LARGE SIGNAL TRANSIENT



PERCENTAGE OVERSHOOT vs. LOAD RESISTANCE (V_{IN} = $100mV_{P-P}$)

